I. $\mathcal{O}_K$ is a subring of K

Definition I.0.1

Let R and S be rings. An element $s \in S$ is called integral over R if it is the root of some monic polynomial in $R[x]$.

Remark I.0.1

This definition is equivalent to say that there is some integer $n > 0$ such that s^n is some R-linear combination of $s^0, s^1, \ldots, s^{n-1}$.

Example I.0.1

Let $x \in \mathbb{Q}$, then x in integral over $\mathbb{Z}$ if and only if $x \in \mathbb{Z}$.

Proof. $\Leftarrow$: Clear.
$\Rightarrow$: If $x = \frac{a}{b}$, $a, b \in \mathbb{Z}$, $\gcd(a,b) = 1$, $b > 0$. Say

$$x^n + c_1 x^{n-1} + \cdots c_n = 0$$

where $c_1, \ldots c_n \in \mathbb{Z}$, then

$$a^n + c_1 a^{n-1} b + \cdots c_n b^n = 0$$

we can move a^n to one side

$$a^n = -c_1 a^{n-1} b - \cdots c_n b^n$$

since b divided RHS, $b \mid a^n$. Since $\gcd(a,b) = 1$, $b \mid 1$ (By Bezout Theorem or other method like prime factorization). So $\frac{a}{b}$ is an integer. $\mathbb{Z}$

Now we make distinction between the following two definitions.

Definition I.0.2

An algebraic number if a root of some polynomial in $\mathbb{Z}[x]$

Definition I.0.3

An algebraic integer if a root of some **monic** polynomial in $\mathbb{Z}[x]$.

Lemma I.0.1

Let M be a finitely generated S-module. Let R be a subring of S such that S is a finitely generated R-module. Then M is a finitely generated R-module.

Proof. Let $m_1, \ldots m_l$ be the generators of M. Let $s_1, \ldots s_k$ be the generators of S. We claim that

$$s_i m_j \quad 1 \le i \le k, 1 \le j \le l$$

generates M.

Given $x \in M$, since M is a finitely generated S-module,

$$x = \sum_{j=1}^{l} x_j m_j \quad x_j \in S$$

For each x_j, we can express it as

$$x_j = \sum_{i=1}^{k} y_{ij} s_i \quad y_{ij} \in R$$

Then

$$\begin{aligned} x &= \sum_{j=1}^{l} (\sum_{i=1}^{k} y_{ij} s_i) m_j \\ &= \sum_{j=1}^{l} \sum_{i=1}^{k} y_{ij} (s_i m_j) \end{aligned}$$

So $s_i m_j$ generates M. $\mathbb{Z}$

Remark I.0.2

If $R \subset S$ as subring. For $x \in S$, write $R[x]$ for all

$$\sum_{i=0}^{n} r_i x^i \quad r_i \in R$$

Lemma I.0.2

Let M be an R-module. If A is some matrix

$$A = (a_{ij})_{1 \le i \le n, 1 \le j \le n} \quad a_{ij} \in R$$

such that

$$A \begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix}$$

for some $m_i \in M$. Then

$$(\det(A)) \begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

Proof. Multiply

$$A \begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix}$$

by the adjoint matrix

$$[(-1)^{i+j} A_{ij}]_{1 \le i \le n, 1 \le j \le n}$$

Then

$$[(-1)^{i+j} A_{ij}]_{1 \le i \le n, 1 \le j \le n} \cdot A \begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix} = \det(A) \begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix}$$

$\mathbb{Z}$

Theorem I.0.3

Let R be a subring of S, let $x \in S$. The following are equivalent.

(1) x integral over R.

(2) $R[x]$ is a finitely generated R-module.

(3) $R[x]$ is contained in a subring T of S, which is a finitely generated R-module.

Proof. (1) $\Rightarrow$ (2): Shown in the previous remark.

(2) $\Rightarrow$ (3): Clear.

(3) $\Rightarrow$ (1): Let $t_1, \dots t_n$ be the generators of T. Since $x \in T$,write

$$xt_j = \sum_{i=1}^{n} r_{ij} t_i \quad r_{ij} \in R$$

We can write down it as a linear transformation.

$$\begin{bmatrix} xt_1 \\ \vdots \\ xt_n \end{bmatrix} = \begin{bmatrix} & & \\ & r_{ij} & \\ & & \end{bmatrix} \cdot \begin{bmatrix} t_1 \\ \vdots \\ t_n \end{bmatrix}$$

So we get

$$(x\mathrm{Id}_n - [r_{ij}]) \begin{bmatrix} t_1 \\ \vdots \\ t_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

By Lemma I.0.2,

$$\det(x\mathrm{Id}_n - [r_{ij}]) \begin{bmatrix} t_1 \\ \vdots \\ t_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

Let $\det(x\mathrm{Id}_n - [r_{ij}]) = D$. Then $Dt_i = 0$ for all t_i. Specifically,

$$D \cdot 1 = 0$$

Since D is a polynomial in x with coefficients in R, we deduce that x in integral over R. $\mathbb{Z}$